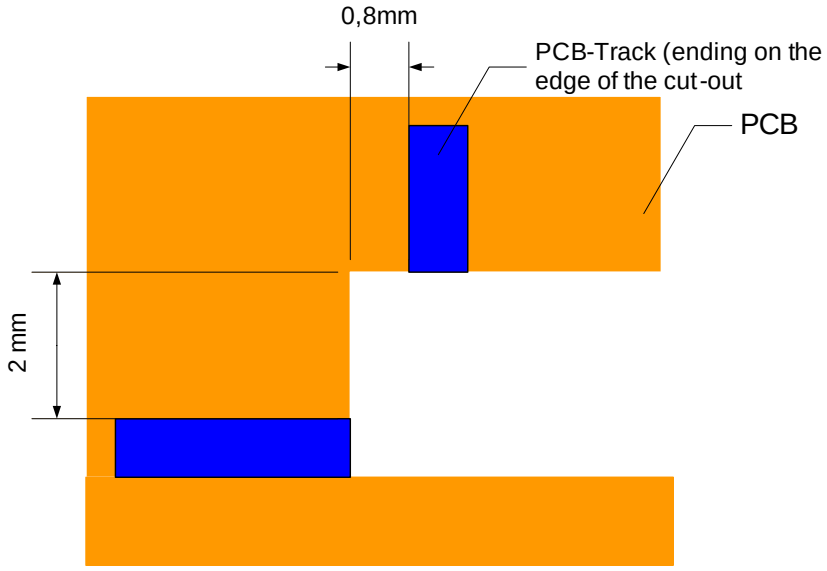
CTL DECISION SHEET

Standard(s):	Subclause(s):	No.	Year
IEC 60664-1/2002 ed1.2 **IEC 60664-1/2007 ed2.0** **All equipment standards based on these**	**4.2** **6.2**	**DSH 0717**	**2008**
Categories: **Various, General**		**Developed by:** **WG2 - WG4**	
Subject: **Creepage and clearance determination, groove, recess, corner, bridging of distance X**	**Key words:** - **Creepage** - **Clearance** - **Groove** - **Recess** - **Corner** - **Bridging of distance X**	**Decision approved at the 2009 CTL Plenary Meeting**	

Questions:

1) How shall the examples given in IEC 60664 be interpreted for intermediate situations?

2) How shall the example given below be judged correctly, based on the examples given in IEC 60664-1 and based on the assumption that X = 1 mm?



3) When shall the sentence:
"If the associated clearance is less than 3 mm, the minimum dimension X may be reduced to one third of this clearance." be applied?
Does this sentence refer to the actual clearance measured or to the minimum clearance required?

Decisions:

Ad 1)
An intermediate situation not covered directly by the Standard shall be determined by considering it as an extension of the most appropriate example from the Standard. An example of the methodology is given in the Explanatory Notes to Question 2."

Ad 2)
The example shall be judged without any bridging of the corner in the cut-out → creepage = 2,8 mm, no shortcut applicable.

Ad 3)
The sentence shall always be applied whenever the required clearance is less than 3 mm and the allowed reduction of the minimum dimension X shall be referred to 1/3 of the required minimum clearance.

Explanatory Notes:

Extract from the standard:

6.2 Measurement of creepage distances and clearances

The dimension *X*, specified in the following examples, has a minimum value depending on the pollution degree as follows:

Pollution degree	Dimension *X* minimum value
1	0,25 mm
2	1,0 mm
3	1,5 mm

If the associated clearance is less than 3 mm, the minimum dimension *X* may be reduced to one-third of this clearance.

The methods of measuring creepage distances and clearances are indicated in the following Examples 1 to 11. These cases do not differentiate between gaps and grooves or between types of insulation.

The following assumptions are made:

- any recess is assumed to be bridged with an insulating link having a length equal to the specified width X and being placed in the most unfavourable position (see Example 3);
- where the distance across a groove is equal to or larger than the specified width X, the creepage distance is measured along the contours of the groove (see Example 2);
- creepage distances and clearances measured between parts which can assume different positions in relation to each other, are measured when these parts are in their most unfavourable position.

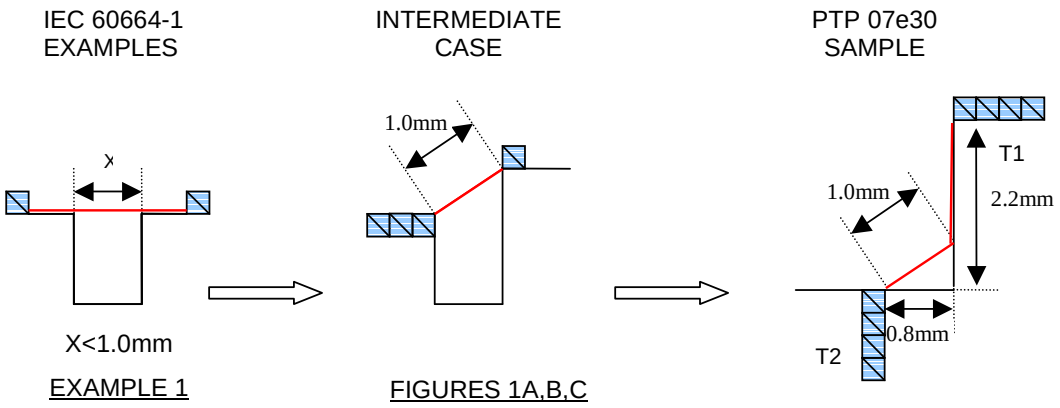
Ad 1) and 2)

Various understandings provided by participants are presented here, but only example 3 is considered as the correct and most appropriate application.

CREAPAGE PATH
ALL CASES DEPICTED FOR PD2

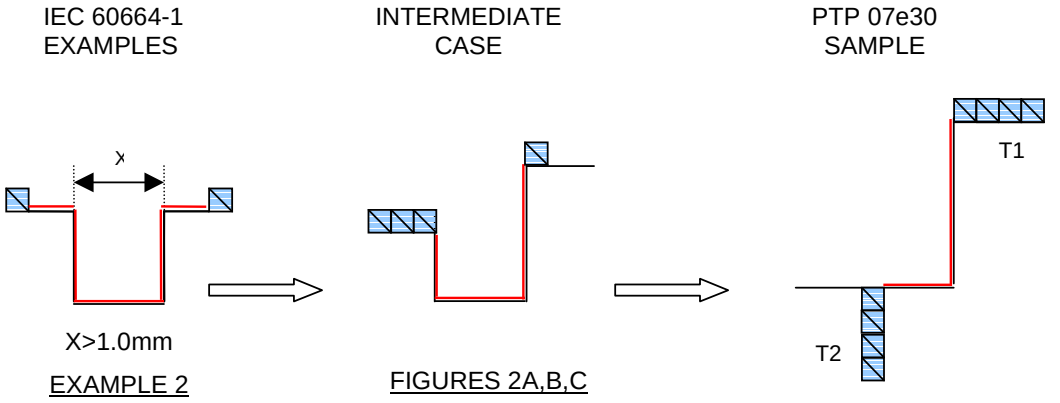
Sample as an Extension of Example 1

The sample could be considered as an extension of Example 1, with the conductors moved to the edges of the rectangular groove and with one side of the groove reduced to zero depth (figure 1). In this interpretation the groove would be bridged by a link x = 1.0mm (for PD2) at an angle determined by the geometry of the groove.



Sample as an Extension of Example 2

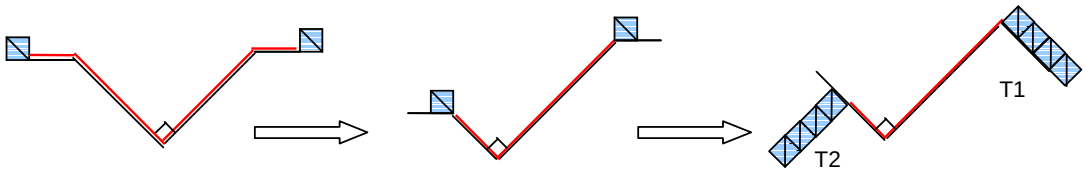
Alternatively, the sample could be considered as an extension of Example 2, with one side of the rectangular groove reduced to zero, and the measurement of x between the shoulders of the groove (figure 2C). Since the groove angle is 90 degrees the corner is not bridged.



Sample as an Extension of Example 3

However, the sample could also be considered as an extension of Example 3, with one of the conductors moved down one side to be close to the bottom of the groove (figure 3C).
Since the groove angle is 90 degrees the corner is not bridged.

IEC 60664-1 EXAMPLES | INTERMEDIATE CASE | PTP 07e30 SAMPLE



80° rule applies

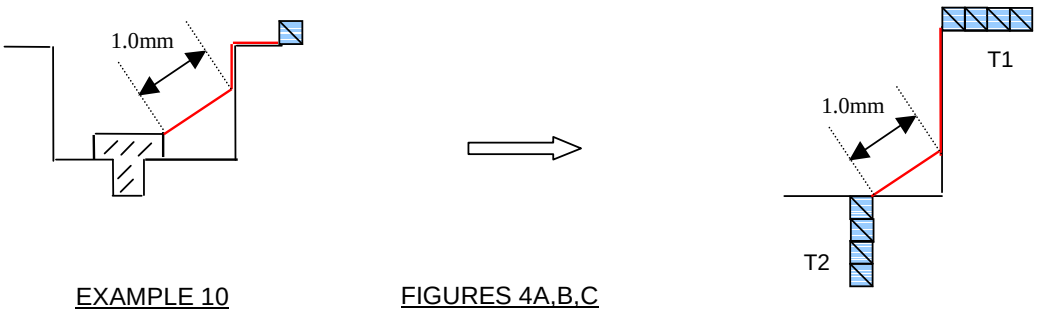
EXAMPLE 3

FIGURES 3A,B,C

Sample as an Extension of Example 10

An argument has been put forward that the rule depicted in Example 10 should apply, since the sample could be considered as similar to that situation with one of the conductors (say T2) taking the place of the screw head and flush with the 'bottom' face of the groove (figure 4C). Applying this rule the corner would be bridged by a link = 1.0mm for PD2, at an angle determined by the geometry.

IEC 60664-1 EXAMPLES | INTERMEDIATE CASE | PTP 07e30 SAMPLE



EXAMPLE 10

FIGURES 4A,B,C

Discussion

The determination of the most reasonable interpretation of the rules for this case depends on what the Standards makers had in mind when creating the rule for bridging a groove. The only guidance given in the Standard (clause 6.2) is:
"The following assumptions are made:

- Any **recess** is assumed to be bridged with an insulating link having a length equal to the specified width x and being placed in the most unfavourable position."

The assumption that is commonly made is that a **recess**, however defined, could attract dust etc and so create a surface for the creepage path.

If the Standards writers had intended **every** gap of 1.0mm created by a change of direction of the surface to be bridged they would not have drawn Examples 1, 2, 4, 6, 7, and 9 the way they did. Therefore it must be concluded that there is a point at which the change in surface direction is no longer regarded as a recess.

Examples 1, 2, 6, 7, and 9 all show 90 degree groove angles which are not treated as recesses. Example 4 shows angles of around 110 degrees which are not treated as recesses.

The CTL Decision 590 effectively defines the boundary (an angle of 80 degrees) when a change of surface direction creates a recess.

The judgment of whether Example 10 can be applied depends on whether a recess is deemed to exist. In example 10 of the Standard, the screw head which is proud of the insulating surface creates a recess. However, in the extension of the example to the sample situation given in the explanatory notes, a recess no longer exists, and all that remains is a right-angled groove. Therefore Example 10 is not an appropriate model for the sample situation given in the explanatory notes.

Of the three remaining models, Example 2 leads to the bridging of a corner where no recess exists. Both Example 2 and Example 3 lead to following the edges of the groove, but the one which most simply corresponds to the example of Question 2 is the V-shaped groove (Example 3) together with CTL decision 590 (80 degree rule).

Ad 3)

1. The practical application of the C/3 reduction when applied for required clearances less than 3mm leads for a great majority of products marketed (rated voltage less than 300 V) to the case that the exception becomes the rule.

2. The C/3 criteria makes no distinction between pollution degrees which leads, for example, to the case that for pollution degree 1 the result is not to increase the measured creepage distance but to reduce it (while for pollution degree 2 or 3 the application leads to an increase of the measured creepage distance).

3. Another point difficult to understand is the relation with other requirement of the standard IEC 60664: Table 2 indicates in note (3) that for all the lower impulse voltages of the standard the minimum clearance required for pollution degree 2 and 3 are limited not by the overvoltage value but this is limited by the “reduced capacity of the associated creepage distance to support voltage in humid conditions”. This limit is 0,2 mm for PD 2 and 0,8 mm for PD 3.

4. If associated clearance is identified with required clearance, for a typical 230 V equipment with overvoltage category II, the required clearance is 1,5 mm for all pollution degrees. The C/3 value is then 0,5 mm. Comparing this with the data of the previous paragraph, we see that for pollution degree II this could be consistent (0,5mm > absolute minimum of 0,2mm), but for Pollution Degree 3, 0,5mm is less than the 0,8mm absolute minimum required.

5. If it is accepted that a clearance shorter than 0,8 mm could be bridged in a PD 3 environment, then it is not logical to accept that a 0,5 mm clearance can break a creepage trajectory.